

# JUNSEOK (JUNE) OH

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## RESEARCH INTERESTS

Computer Vision, Large Language Models, Multimodal Social Attention: how an agent infers who needs attention, what matters in the moment, and when assistance is appropriate, using cues such as gaze, head orientation, language, and turn-taking.

## EDUCATION

### University of Pennsylvania

*MSE in Data Science*

Philadelphia, PA

### Purdue University

*BSc in Computer Science, Minor in Mathematics*

West Lafayette, IN

## PUBLICATIONS

- [1] D. W. Lee, **J. Oh**, S. Gillet, C. Ahuja, A. Tagliabue, Y. Kim, L.-P. Morency, C. Breazeal, H. W. Park. "Real Time Open-Vocabulary Referential Gaze Control for Listening Robots." *In Submission* 2027.
- [2] H. Min\*, H. A. Mina, **J. Oh**, A. J. Deering, J. P. Robinson, B. Rajwa, E. Bae. "Smartphone-Integrated Optomechanical Dual-Mode Instrument for Salmonella Typhimurium Detection." *IEEE Sensors Journal* 2025.

## RESEARCH EXPERIENCE

### MIT Media Lab, Personal Robots Group

Feb 2025 – Present

*Advisors: Dong Won Lee, Dr. Hae Won Park, Prof. Cynthia Breazeal, Prof. Louis-Philippe Morency (Remote) Cambridge, MA*  
*Real-time referential gaze control*

- Second author on [1]: low-latency, open-vocabulary referential gaze control for listening robots, combining a multimodal foundation model and object segmentation with privileged information distillation.
- Curated and audited a paired 360° video evaluation of explicit and implicit spoken references to the same targets (259 targets, 518 references, 79 source videos), and quantified its overlap with training annotations across 131 object groups.
- Built reproducible Python/FFmpeg workflows that render panoramic video frames into the figures of [1], linking source frames, projection geometry, and field-of-view overlays to editable assets.

### *Egocentric head-gaze prediction*

- Curated a 40+ hour conversation-centric egocentric dataset from Aria, Ego4D, and EgoCom, with evaluation principles and ground-truth interaction labels.
- Developed a preprocessing pipeline generating proxy head-gaze labels with CoTracker (validated at 0.47° MAE against IMU) and speaker diarization with WhisperX.
- Implemented a PyTorch framework for distributed multi-GPU training and evaluation over 5M+ frames of video.
- Created evaluation benchmarks with heuristic behavioral baselines for social dynamics and gaze control.

### Purdue University, Department of Mechanical Engineering

Sep 2022 – Oct 2023

*Advisors: Hyun Jung Min, Euiwon Bae*

*West Lafayette, IN*

- Developed a real-time Android application for bacterial detection [2], integrating OpenCV image recognition with a quartz crystal microbalance (QCM) sensor.
- Implemented a pipeline that logs experimental data and plots frequency and temperature changes in real time.
- Improved detection by automating image capture on sensor-threshold triggers.

### Stanford Cardiovascular Institute

Jan – Aug 2022, Jan – May 2024

*Advisor: Siyeon Rhee*

*(Remote) Stanford, CA*

- Built an automated scRNA-seq analysis pipeline over public GEO datasets, identifying differentially expressed genes between adult and p12 developmental stages.
- Performed gene set enrichment analysis (GSEA) and cell-to-cell interaction profiling to identify markers in adipogenesis and angiogenesis.
- Integrated epicardial adipose tissue (EAT) datasets across sources, and visualized single-cell data with PCA and UMAP.

## Wittgen Biotechnology, UC Berkeley SkyDeck

May 2022 – Aug 2022

Research Intern

(Remote) Berkeley, CA

- Applied machine learning to RNA sequencing data from 40+ cancer patients for high-resolution tumor classification and tailored drug recommendation.
- Differentiated cell types in single-cell RNA analysis with Seurat, and collected and curated patient data for an AI platform profiling cancer heterogeneity.

## PROFESSIONAL EXPERIENCE

### Google LLC

Mar 2026 – Present

Software Engineer, Search Ads Google Experience

Mountain View, CA

- Core engineer on the Universal Commerce Protocol (UCP), Google's open standard for agentic commerce, and Direct Offers, a Google Ads pilot that presents exclusive offers to shoppers in AI Mode.
- Own **conversion attribution for UCP**, linking UCP purchases to the ad interactions that drove them.
- Build **Direct Offers** across ad surfaces, including per-surface backend logic, the frontend, and core algorithms.
- Develop high-concurrency backend services and data-fetching infrastructure for ads delivery, converting sequential query paths into parallel execution across multi-service production systems.
- Scale and maintain high-throughput ingestion pipelines for large-scale ad conversion-event data.

### Hewlett Packard Enterprise, Aruba AI Ops

Jul 2024 – Mar 2026

Cloud Software Engineer

San Jose, CA

- Designed and built an agentic AI system with retrieval-augmented generation (RAG) to automate network root cause analysis from internal documentation.
- Deployed generative AI into production network troubleshooting workflows, improving anomaly detection and recommendation accuracy.
- Optimized DNS and DHCP data-recall algorithms in the AI Ops pipeline, achieving **83%** and **91%** reductions.

### Hewlett Packard Enterprise, Aruba AI Ops

May 2023 – Dec 2023

Data Science Intern

San Jose, CA

- Raised root cause prediction accuracy from **50% to 78%** with a tuned tree-based model.
- Developed and deployed a production ML system integrating Llama-2-7b in a scalable pipeline.
- Engineered 38 features from 2,500+ TAC cases and analyzed 1M+ firewall and DNS logs with PySpark.

## TEACHING

### Purdue University, Department of Computer Science

Jan 2023 – May 2023

Teaching Assistant, CS 25200 Systems Programming (Instructor: Gustavo Rodriguez-Rivera)

West Lafayette, IN

- Mentored 100+ students weekly on process control, concurrency (pthreads), and memory management.
- Guided C/C++ debugging with GDB and Valgrind for segmentation faults, memory leaks, and race conditions.
- Led lab sessions on shell scripting and network programming (TCP/IP sockets).

## MILITARY SERVICE

### Republic of Korea Army

2019 – 2021

Sergeant, honorably discharged

## TECHNICAL SKILLS

|                           |                                                                               |
|---------------------------|-------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, SQL, Java, C/C++                                                      |
| <b>ML &amp; Data</b>      | PyTorch, TensorFlow, Spark (PySpark), Scikit-learn, NumPy, Pandas, LLMs & RAG |
| <b>Vision &amp; Media</b> | OpenCV, FFmpeg, CoTracker, WhisperX                                           |
| <b>Infrastructure</b>     | AWS, GCP, Kubernetes, Kafka, Git                                              |